

LEONARDO AVONI



PHD CANDIDATE
FLIGHT MECHANICS & AERODYNAMICS

ABOUT

Mechanical Engineer (EPFL) and PhD Candidate (ISAE-Supaero/ENAC) specializing in aerodynamics and control. Skilled in CAD and programming (Python, MATLAB), with hands-on engineering experience.

CONTACT

- ✉ avonileonardo@gmail.com
- 📍 Toulouse, France
Geneva, Switzerland
- 🌐 [linkedin.com/in/leonardo-avoni](https://www.linkedin.com/in/leonardo-avoni)
- 🌐 leonardoavoni.com

LANGUAGES

- **Italian, French:** Mother tongue
- **English:** Fluent

TECHNICAL SKILLS

Aerodynamics

- ASWING, AVL
- ANSYS Fluent

CAD (Computer-Aided Design)

- CATIA V5, SketchUp
- Fusion 360, OnShape
- ReCap and Inventor

UAS (Uncrewed Aerial Systems)

- PX4, QGroundControl
- Gazebo, MAVSDK
- UAS assembly and flight

Coding

- Fortran, Python, C, Labview
- MATLAB + Simulink
- HTML/CSS, LaTeX/Overleaf
- Git, GitHub, VSCode, Ubuntu

Hardware

- 3D printing (FDM / SLA)
- Bike Maintenance
- Hose renovation
- Basic woodworking

CERTIFICATIONS

- OPEN A1/A3 Drone Pilot License
- PSC — First Aid
- PSSM — Mental Health First Aid
- Cat. B Driving License

EDUCATION

- **ISAE-Supaero & ENAC, France** **2023 - Present**
PhD, Flexible Aircraft Dynamics and Control
 - Directed by Prof. Jean-Marc Moschetta & Prof. Jean-Philippe Condomines
 - Aeroelastic design and control of flexible-wing aircraft
 - Student representative for PhD candidates, MEGeP / EDyF school
- **EPFL, Switzerland** **2021 - 2023**
Master, Mechanical Engineering
 - Specialization: Fluid Mechanics, Minor: Space Technologies
 - Teaching Assistant position in BsC and MsC Fluid Mechanics Course
- **EPFL, Switzerland** **2018 - 2021**
Bachelor, Mechanical Engineering
 - Strong foundation mainly in mathematics, thermodynamics, mechanics, and fluid mechanics. Additional courses in electronics and programming
- **Ferney-Voltaire International School, France** **2015 - 2018**
 - Scientific Baccalauréat; Highest Honors (Mention Très Bien)
 - Group project presented at CERN (TPE / Mon TPE en 15', 2018)

EXPERIENCE & RELEVANT PROJECTS

- **PhD Contributions, ISAE-Supaero/ENAC, France** **2023 - Present**
Current contributions revolve around extensions of ASWING: a fast, Fortran-written, flexible aircraft flight dynamics software from MIT
 - WingLoop development: a tool combining **Python/Matlab-written control** laws with ASWING for time-efficient closed-loop flight dynamics (AIAA Aviation 2025, DOI: 10.2514/6.2025-3425)
 - Wrote the ASWING Extended User Manual, featured on the official MIT ASWING website
 - Implemented code for Wingshape, OptiTrack, load cell and servomotor communication for wind tunnel testing
 - Working on ASWING stall implementation for aircraft deep stall analysis
 - 5-month **internship supervisor**, for **flexible aircraft LQR control**
- **Drone building: theory and practice, Personal Project** **2026 - Present**
Personal project to learn **UAV design**, build, simulation and flight:
 - Literature review, need identification, **component selection**, **wiring scheme** for PX4-based 5" quadcopter
 - Drone simulator, **SITL PX4 + QGroundControl + Gazebo** custom Ubuntu setup, including 5" drone model and realistic Gazebo world.
 - Python MAVSDK experimentation within SITL setup
 - **Real life quadcopter build and flight**. 3D-printed parts were used, **DJI FPV video equipment** was then added

EXPERIENCE & RELEVANT PROJECTS (CONTINUED)

Master Project: DroneDetective, EPFL/Elythor, Switzerland

Spring 2023

Project within the Laboratory of Intelligent Systems (LIS) of Prof. Floreano, supervised by LIS-spinoff Elythor. The project involved the developed of a Python software, called **DroneDetective**, for the automation of wind tunnel testing of morphing VTOL drones.

- DroneDetective **coordinated data between a Robotic arm, a wind tunnel, the load cell, a MoCap system and the drone**, through respective API's.
- A GUI was also built for easier operator monitoring
- Procedures for **center-of-mass coordinates extraction**, aerodynamic characterization and **vibration analysis** were coded

Final Grade: 5.75/6.0

CFD Internship, EPFL, LIS, Switzerland

2023

Analysis on **wing and propeller interactions** using Ansys Fluent. The project included:

- Literature review, specifically for subsonic UAVs, then general guidelines for wing alone, propeller alone and wing+propeller setup
- Generalization for **Moving Reference Frame and Sliding Mesh** approaches
- Creation of **video tutorials for the Ansys Fluent** setup to use

Industrial Internships, Synova SA, Switzerland

Summer 2021 and 2022

- 2021: Improved 5-axis CNC calibration using state-of-the-art ISO-standard techniques. First, the weak points of the initial calibration techniques were identified, then an **improvement plan was proposed, implemented and validated on the LCS-305 machine**. The plan was designed according to he possibilities offered by **Bosch CNC's**
- 2022: Analyzed the influence of stabilization gases on water jet stability for **Laser MicroJet® Technology**. Theoretical and practical **analysis were done on 3 and 5 axis machines**. The internship also included machining works for external aeronautical clients involving deep cutting.

Project: Supersonic Wind Tunnel, EPFL/HEPIA, Switzerland

Spring 2022

- Designed and tested a **small-scale open wind tunnel for supersonic flow**. An air compressor was used as high-pressure reservoir.
- Nozzle design was created according to compressible flow mechanics
- The nozzle was then **SLA 3D printed**, and tested. Flow observations were done with a **Schlieren observation bench**

OTHER SKILLS

- Road Cycling, Dancing, Running.
- Experience in Climbing, Judo, and Sailing.
- Classical Guitar (Intermediate).
- Electric Guitar (Basic).
- Music Theory (Basic)